

FRAUD DETECTION PIPELINE

Implementation Report

DecodeLabs Data Science Industrial Training

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Project: Project 2 – Supervised Learning (Fraud Detection Pipeline)

Dataset: Synthetic Credit Card Fraud Detection Dataset

1. Introduction

Fraud detection is a binary classification problem in which machine learning models distinguish between legitimate and potentially fraudulent transactions.

The main challenge in fraud detection is class imbalance, because fraudulent transactions are generally much less common than legitimate transactions. According to the DecodeLabs Project 2 requirements, this project focuses on handling class imbalance using SMOTE (Synthetic Minority Over-sampling Technique) and training classification models such as Logistic Regression and Random Forest. The models are evaluated using Precision, Recall, and ROC-AUC rather than relying primarily on Accuracy.

The objective of this implementation is to develop a complete fraud detection pipeline, compare multiple supervised learning algorithms, tune the Random Forest model, and evaluate the models using appropriate metrics for an imbalanced classification problem.

2. Project Objectives

The main objectives of this project are:

- To understand and analyze an imbalanced fraud detection dataset.
- To preprocess numerical and categorical transaction features.
- To handle class imbalance using SMOTE.
- To implement Logistic Regression as a baseline classification model.

- To implement Random Forest as a nonlinear classification model.
- To evaluate the models using Precision, Recall, F1-score, and ROC-AUC.
- To perform hyperparameter tuning for the Random Forest model.
- To compare the performance of the different models.
- To identify the most suitable model for the given fraud detection dataset.

These objectives directly follow the DecodeLabs Project 2 requirements, which specify SMOTE, Logistic Regression, Random Forest, and evaluation using Precision, Recall, and ROC-AUC.

3. Dataset Description

The dataset used in this project is a **synthetic Credit Card Fraud Detection Dataset** containing 10,000 transaction records and 10 columns.

The target variable is **is_fraud**, which indicates whether a transaction is fraudulent or legitimate.

After inspecting the actual uploaded CSV file, the dataset contained:

Property	Value
Total records	10,000
Total columns	10
Fraudulent transactions	151
Legitimate transactions	9,849
Observed fraud rate	1.51%
Missing values	0
Duplicate rows	0
Target variable	is_fraud

The dataset is therefore highly imbalanced, with legitimate transactions representing the large majority of observations.

3.1 Features

The dataset contains the following main transaction features:

- **transaction_id** – unique identifier for each transaction.
- **amount** – monetary value of the transaction.
- **transaction_hour** – hour at which the transaction occurred.
- **merchant_category** – category of the merchant.
- **foreign_transaction** – indicator showing whether the transaction was foreign.
- **location_mismatch** – indicator of a mismatch between transaction location and expected location.
- **device_trust_score** – score representing the trust level of the device.
- **velocity_last_24h** – transaction activity or velocity during the previous 24 hours.
- **cardholder_age** – age of the cardholder.
- **is_fraud** – target variable indicating fraudulent or legitimate transaction.

The `transaction_id` column was excluded from model training because it is an identifier rather than a meaningful predictive feature.

4. Data Preprocessing

Before model training, the dataset was inspected for data quality issues.

The following checks were performed:

- Dataset dimensions were examined.
- Column names and data types were inspected.
- Missing values were checked.
- Duplicate records were checked.
- The distribution of the target variable was analyzed.
- Numerical feature distributions were explored.
- Categorical feature distributions were examined.

The dataset contained **no missing values and no duplicate rows**.

4.1 Feature Preparation

The target variable is `_fraud` was separated from the predictor variables.

The `transaction_id` column was removed because it represents an identifier and does not provide meaningful information about whether a transaction is fraudulent.

The numerical features were standardized using **StandardScaler**.

The categorical feature `merchant_category` was converted into numerical representation using **One-Hot Encoding**.

5. Train-Test Split

The dataset was divided into training and testing subsets using an **80/20 split**.

Stratification was used during the split to preserve the proportion of fraudulent and legitimate transactions in both the training and testing datasets.

A fixed random state was also used to make the experiment reproducible.

The test dataset was kept separate from the SMOTE process so that synthetic samples were generated only from the training data.

6. Handling Class Imbalance Using SMOTE

One of the most important parts of this project is handling the imbalance between legitimate and fraudulent transactions.

The DecodeLabs project specifically requires the implementation of **SMOTE (Synthetic Minority Over-sampling Technique)** to address this problem.

SMOTE increases the representation of the minority class by generating synthetic observations based on existing minority-class samples.

Instead of simply duplicating fraudulent transactions, SMOTE creates new synthetic samples that are based on the characteristics of existing minority observations.

In this project, SMOTE was incorporated into the machine-learning pipeline so that the oversampling was performed during training rather than before the train-test split.

This helps prevent information from the test dataset from influencing the training process.

7. Machine Learning Models

Two supervised classification algorithms were implemented.

7.1 Logistic Regression

Logistic Regression was used as a baseline classification algorithm.

It is a linear classification method that estimates the probability of an observation belonging to a particular class.

In this project, Logistic Regression provides a useful baseline against which the more complex Random Forest model can be compared.

7.2 Random Forest

Random Forest is an ensemble machine-learning algorithm that combines multiple decision trees.

It can capture nonlinear relationships between features and the target variable and is therefore suitable for classification problems where the relationship between transaction characteristics and fraud may not be purely linear.

The Random Forest model was trained using the same preprocessing and SMOTE-based pipeline.

8. Model Evaluation

Because the dataset is imbalanced, **Accuracy was not considered the primary evaluation metric.**

A model could achieve high Accuracy simply by predicting most transactions as legitimate while failing to identify fraudulent transactions.

Therefore, the following metrics were used:

Precision

Precision measures the proportion of transactions predicted as fraudulent that were actually fraudulent.

A higher Precision means fewer legitimate transactions are incorrectly flagged as fraud.

Recall

Recall measures the proportion of actual fraudulent transactions that were successfully detected.

A higher Recall means the model is better at finding fraudulent transactions.

F1-Score

F1-score combines Precision and Recall into a single metric and is useful when both false positives and false negatives are important.

ROC-AUC

ROC-AUC measures the model's ability to distinguish between the positive and negative classes across different classification thresholds.

The DecodeLabs project specifically emphasizes Precision, Recall, and ROC-AUC instead of relying on Accuracy for this imbalanced fraud-detection problem.

9. Model Results

The following results were obtained from the implementation:

Model		Precision	Recall	F1-Score	ROC-AUC
Logistic Regression		0.287	0.967	0.443	0.993
Random Forest		1.000	0.633	0.776	0.999
Tuned Forest	Random	1.000	0.633	0.776	0.999

10. Results Analysis

10.1 Logistic Regression

Logistic Regression achieved a **Recall of 0.967**, which means it detected almost all of the fraudulent transactions in the test set.

However, its **precision was only 0.287**.

This indicates that although the model was very effective at finding fraudulent transactions, it also produced a relatively high number of false-positive alerts.

Therefore, Logistic Regression was highly sensitive to fraudulent transactions but less precise in deciding which transactions should actually be classified as fraud.

10.2 Random Forest

Random Forest achieved:

- **Precision = 1.000**
- **Recall = 0.633**
- **F1-score = 0.776**
- **ROC-AUC = 0.999**

The model detected **19 out of 30 fraudulent transactions** in the held-out test set.

Its Precision of 1.000 means that none of the transactions predicted as fraudulent in this test split were legitimate transactions.

However, its Recall of 0.633 also shows that the model missed some fraudulent transactions.

Therefore, Random Forest provided a much stronger Precision and overall balance than Logistic Regression, although Logistic Regression achieved higher Recall.

11. Hyperparameter Tuning

Hyperparameter tuning was performed on the Random Forest model using **GridSearchCV**.

The following parameters were explored:

- Number of estimators
- Maximum tree depth
- Minimum samples required to split a node
- Minimum samples required at a leaf node

ROC-AUC was used as the scoring metric during the tuning process.

The selected Random Forest configuration was:

- **n_estimators:** 200
- **max_depth:** None
- **min_samples_split:** 2
- **min_samples_leaf:** 1

The tuned Random Forest produced the same test-set metrics as the original Random Forest in this particular experiment.

This indicates that the original Random Forest configuration was already performing very strongly on the selected dataset.

12. Comparison of Models

The comparison demonstrates an important trade-off between Precision and Recall.

Logistic Regression prioritized Recall:

Recall = 0.967

This means it was highly effective at detecting fraudulent transactions, but it generated more false-positive alerts.

Random Forest prioritized Precision:

Precision = 1.000

This means the transactions it classified as fraudulent were highly reliable, but it failed to detect some fraudulent transactions.

Random Forest also achieved the highest ROC-AUC:

$$\text{ROC-AUC} = 0.999$$

Based on the overall results from this experiment, **Random Forest was selected as the stronger overall model**, particularly because of its excellent Precision, F1-score, and ROC-AUC.

However, the choice of model in a real fraud-detection system would depend on the relative cost of false positives and false negatives.

13. Important Data Quality Observation

During the dataset analysis, an important discrepancy was identified.

The dataset description indicated an expected fraud percentage of approximately 4–5%. However, inspection of the actual uploaded CSV showed:

- 151 fraudulent transactions
- 9,849 legitimate transactions
- 1.51% observed fraud rate

Therefore, the actual CSV values were used for all analysis and model evaluation in this project.

This demonstrates the importance of checking the actual dataset rather than relying only on the dataset description.

14. Limitations

There are several limitations to this project.

14.1 Synthetic Dataset

The dataset used in this project is synthetic rather than real banking transaction data.

Therefore, the model results should **not** be interpreted as evidence that the same performance would be achieved in a real-world financial institution.

Synthetic datasets may contain simplified or artificially strong relationships between features and the target variable.

14.2 Dataset Size

The dataset contains 10,000 transactions, which is useful for a learning project but relatively small compared with the volume of transactions processed by real financial systems.

14.3 Model Threshold

The evaluation was performed using the standard classification threshold.

Future work could investigate different probability thresholds to find a better balance between fraud detection and false-positive alerts.

14.4 Categorical Feature Handling

The dataset contains a categorical feature, `merchant_category`.

A future version could investigate a categorical-aware oversampling approach such as **SMOTENC** to handle categorical variables more appropriately during synthetic sampling.

15. Future Improvements

Several improvements could be explored in future versions of this project:

- Experiment with different classification thresholds.
- Compare additional classification algorithms.
- Investigate SMOTENC for categorical variables.
- Perform more extensive hyperparameter optimization.
- Use precision-recall curves in addition to ROC curves.
- Apply cost-sensitive learning.
- Test the models on a larger and more realistic dataset.
- Perform cross-validation with additional evaluation metrics.

- Analyze feature importance in greater depth.
- Investigate false-positive and false-negative transactions individually.

These improvements could provide a more comprehensive understanding of model behavior in fraud detection.

16. Technologies Used

The following technologies and libraries were used:

- **Python**
- **Pandas**
- **NumPy**
- **Matplotlib**
- **Scikit-learn**
- **imbalanced-learn**
- **Google Colab / Jupyter Notebook**

17. Conclusion

This project implemented a complete supervised machine-learning pipeline for credit-card fraud detection using an imbalanced dataset.

The workflow included data inspection, preprocessing, train-test splitting, SMOTE-based class balancing, Logistic Regression, Random Forest, model evaluation, and hyperparameter tuning.

The results demonstrated that different models can have very different Precision and Recall characteristics.

Logistic Regression achieved very high Recall, making it effective at identifying fraudulent transactions, but it produced more false-positive alerts.

Random Forest achieved excellent Precision and ROC-AUC and provided the strongest overall performance in this experiment.

The project also demonstrated an important machine-learning principle: **a model should not be judged using Accuracy alone when dealing with highly imbalanced data.**

Precision, Recall, F1-score, and ROC-AUC provide more meaningful information about the performance of a fraud-detection model.

Overall, this project provided practical experience in supervised classification, imbalanced-data handling, model comparison, evaluation, and hyperparameter tuning, which are important skills for real-world machine-learning and data-science applications.

18. Reference

DecodeLabs Data Science Industrial Training — Project 2: Supervised Learning (Fraud Detection Pipeline).

The project brief specifies the use of SMOTE for class imbalance, Logistic Regression and Random Forest classification models, and evaluation using Precision, Recall, and ROC-AUC.

Dataset: Synthetic Credit Card Fraud Detection Dataset used for the implementation of this project.